

**Dataset of Recommended 2025 CDF Projects, Loan Applicants,
Empowerment Grants and Bursary Applications from Gwembe District,
Zambia**

Data Description Paper

Abstract

This article describes five structured datasets prepared from the 2025 list of recommended Constituency Development Fund (CDF) projects and applications published by Gwembe Town Council, Zambia. The source document contains community projects, loan applicants, empowerment grant applications, secondary boarding school bursary applications, and skills development bursary applications. The source PDF was processed with Python. Tables were extracted page by page with pdfplumber, cleaned, grouped into five pandas DataFrames, and exported as pipe-delimited (|) CSV data. The final cleaning workflow removed serial-number columns, header rows, total rows, and records with missing values according to dataset-specific rules. It also converted selected financial fields to numeric values and normalized text to lowercase. The resulting in-memory datasets contain 7 community projects, 17 loan applicants, 90 empowerment grant applications, 44 school bursary applications, and 177 skills development bursary applications. The five datasets describe development projects, business and loan applications, empowerment activities, secondary education support, and vocational or skills training support. They can be reused for computational analysis of the geographical distribution and characteristics of recommended CDF interventions in Gwembe District.

Keywords: Constituency Development Fund; Gwembe District; Zambia; public-sector data; PDF table extraction; pandas; pdfplumber; bursaries; empowerment grants

Specifications Table

Field	Description
Subject area	Computer Science / Data Science
Specific subject area	Structured extraction and cleaning of public-sector development data
Type of data	Tabular data
How the data were acquired	Programmatic retrieval of a Gwembe Town Council webpage and extraction of its linked PDF
Data format	Pipe-delimited CSV ()
Data source location	Gwembe District, Zambia
Source period	2025 recommended CDF projects and applications
Number of final datasets	Five
Final retained records	7 community projects; 17 loan applicants; 90 empowerment grants; 44 school bursaries; 177 skills development bursaries
Data processing	PDF table extraction, table grouping, row/column cleaning, missing-value filtering, text normalization, selected numeric conversion, CSV export

Value of the Data

- The data provide a machine-readable representation of five categories of recommended 2025 CDF interventions published for Gwembe District.
- The geographical fields support comparisons across wards, zones and project or school locations.
- The loan and empowerment records provide structured information on business activities, sectors, requested or approved amounts, and applicant locations.
- The education datasets contain information that can support analysis of pupils, students, programmes, institutions, programme duration, sex, and fee requirements.
- The pipe-delimited format can be loaded directly into Python, R, spreadsheet applications, database systems, and other data-analysis tools.

1. Background

Gwembe Town Council published a 2025 document titled “List of Recommended 2025 CDF Projects by CDFC”. The document brings together several categories of recommended activities and applications under the Constituency Development Fund. The community-project section lists seven projects, including health facilities, classroom construction, road maintenance, rural electrification and water-related infrastructure. The source reports a total of K18,304,493.20 for the community-project section.

The same document contains recommended loan applications, empowerment grants, secondary boarding school bursaries and skills development bursaries. The loan section includes business names, wards, zones, contact persons, business activities, sectors, requested amounts and remarks. The source lists a total requested amount of K3,599,000.00 for the loan section.

The empowerment section contains 100 numbered applications in the source and reports total approved funding of K2,259,375.00. Examples of supported activities include goat rearing, poultry, piggery, irrigation, cattle production and bee keeping.

The school-bursary section contains 48 numbered applications and reports total termly fees of K48,000.00 and annual fees of K144,000.00. The source records pupil names, wards, sex, grades, schools, school locations and fee amounts.

The skills-development section records student names, wards, sex, dates of birth, training programmes, training levels, programme duration, training institutions and term tuition amounts. The PDF contains values split across lines in some cells, which is one reason the extracted data require cleaning before computational use.

2. Data Description

The notebook separates the source into five logical datasets. The initial extracted DataFrame dimensions were 8×8 for community projects, 22×10 for loan applicants, 97×5 for empowerment grants, 50×9 for school bursaries and 275×13 for skills development bursaries. These dimensions include source headers, total rows and extraction artefacts that are subsequently handled by the cleaning workflow.

2.1 Community Projects Dataset

The final community-project dataset contains 7 records and 7 variables. The serial-number field was removed, and the source total row was removed. The remaining fields are project name, project description, sector, project type, ward, zone and location. Text values were converted to lowercase before export.

Variable	Description
project_name	Name of the proposed project
project_description	Description of the proposed activity
sector	Sector associated with the project
project	Project type recorded in the source
ward	Ward associated with the project
zone	Zone associated with the project
location	Project site or location

2.2 Loan Applicants Dataset

The final loan dataset contains 17 records and 9 variables. The application-number field was removed. The first two rows were removed because they were repeated header rows, and three records containing missing values were removed by a complete-row missing-value filter. Text fields were converted to lowercase, the requested-loan field was converted to numeric after comma separators were removed, and column names were normalized to lowercase with spaces replaced by underscores. The source contains at least one phone value recorded as 0, which the cleaning workflow does not independently correct.

Variable	Description
name_of_group/name_of_business/company	Applicant, group, business or company name
ward	Ward of the applicant
zone	Zone associated with the applicant
contact_person_(name)	Contact person's name
contact_person_(phone_no.)	Contact telephone number
type_of_business	Business activity
sector	Economic or business sector

loan_amount_requested	Requested loan amount
remarks	Additional application information

2.3 Empowerment Grants Dataset

The final empowerment-grants DataFrame contains 90 records and 4 variables. The serial-number field was removed. The first row was removed, and records containing missing values were then removed. The approved-amount field was converted to numeric after comma separators were removed. Column names were normalized to lowercase with spaces replaced by underscores, and text values were converted to lowercase.

Variable	Description
name_of_applicant	Applicant, group, club or cooperative name
ward	Ward associated with the applicant
project_title	Proposed empowerment activity
approved_amount	Approved amount recorded for the application

2.4 Secondary Boarding School Bursaries Dataset

The final school-bursary DataFrame contains 44 records and 8 variables. The serial-number field and first row were removed. One sex value was manually set to M in the notebook before missing-value filtering. Rows still containing missing values were then removed. Text fields were converted to lowercase, column names were normalized, and termly and annual fee fields were converted to numeric values after comma separators were removed.

Variable	Description
name_of_pupil	Name of pupil
ward	Ward of the pupil
sex_(m/f)	Sex recorded for the pupil
grade	Grade
name_of_school	School attended
school_location_(district)	District/location of school
termly_fees	Fees per term
annual_fees	Annual fee requirement

2.5 Skills Development Bursaries Dataset

The final skills-development DataFrame contains 177 records and 12 variables. The serial-number field and first row were removed. Text values were converted to lowercase and column names were normalized. The

notebook then applied three sequential missing-value filters: records without a skill or programme were removed; records without a student name, ward or date of birth were removed; and records without values for Term One, Term Two or Term Three were removed. Unlike the loan, empowerment and school datasets, the final notebook does not convert the skills financial fields to numeric values or parse dates into a date type.

Variable	Description
name_of_student	Name of student
ward	Ward of the student
sex_(m/f)	Sex recorded in the source
date_of_birth	Date of birth as extracted
name_of_skill/programme	Skill or training programme
level_of_skill	Training level, such as certificate or diploma
programme_duration_(no._of_months)	Programme duration as recorded
training_institute_(tevet/zns)	Training institution
term_one	Term One tuition amount
term_two	Term Two tuition amount
term_three	Term Three tuition amount
annual_tuition_fees	Annual tuition amount

3. Experimental Design, Materials and Methods

3.1 Data Acquisition

The notebook uses Python requests to retrieve the Gwembe Town Council webpage and BeautifulSoup to identify links containing PDF files. The first identified PDF is downloaded and saved locally. The extraction stage then operates on the local Gwembe PDF. The source document contains 35 pages and the notebook's table extraction procedure produces 36 page-level table objects.

3.2 PDF Table Extraction

The PDF is processed one page at a time with pdfplumber. The notebook uses line-based vertical and horizontal extraction strategies with snap_tolerance=7, join_tolerance=7 and edge_min_length=3. The extracted tables are inspected before being combined. A helper function removes line breaks and repeated spaces from individual cells and another helper function pads or truncates rows so that they match the expected column count.

3.3 Dataset Construction

The 36 extracted tables are grouped by their position in the source document: table 0 is assigned to community projects; tables 1–3 to loan applicants; tables 4–9 to empowerment grants; tables 10–15 to school bursaries; and tables 16–35 to skills development bursaries. Each group is converted into a pandas DataFrame using a predefined column list.

Extracted tables	Dataset	Initial dimensions
0:1	Community projects	8×8
1:4	Loan applicants	22×10
4:10	Empowerment grants	97×5
10:16	School bursaries	50×9
16:36	Skills development bursaries	275×13

3.4 Cleaning Operations

The cleaning operations are dataset-specific rather than one common transformation applied to all records. The community dataset removes its serial-number column and row 7, which is the source total row. The loan dataset removes the application-number column, removes the first two rows, applies dropna to remove incomplete records, lowercases the data, converts the requested amount to numeric, and normalizes column names. The empowerment dataset removes the serial-number column and first row, lowercases the data, converts approved amounts to numeric, removes incomplete records, and normalizes column names. The school dataset removes the serial-number column and first row, applies the notebook's manual sex correction, removes incomplete rows, lowercases values, normalizes column names and converts the two fee fields to numeric. The skills dataset removes the serial-number column and first row, lowercases values, normalizes column names, and applies the three sequential missing-value filters described above.

3.5 Records Removed During Cleaning

The cleaning workflow substantially reduces the extracted tables. The final retained counts and the main removal operations are shown below.

Dataset	Initial rows	Final rows	Main reductions
Community projects	8	7	Removed S/N; removed total row
Loan applicants	22	17	Removed No.; removed 2 header rows; removed 3 incomplete rows
Empowerment grants	97	90	Removed S/N; removed first row; removed rows with missing values
School bursaries	50	44	Removed S/N; removed first row; manual sex correction; removed incomplete rows
Skills development bursaries	275	177	Removed S/N; removed first row; sequential missing-value filters

3.6 Financial and Text Fields

The notebook converts Loan Amount Requested, Approved Amount, Termly Fees and Annual Fees to numeric values after removing comma separators. The skills-development tuition fields are not converted to numeric values in the final notebook. Text values in all five cleaned datasets are lowercased before export, while column names are normalized in the datasets where the notebook explicitly performs column normalization.

4. Data Validation and Quality

The final notebook performs missing-value inspection and removes records according to the rules described above. It also checks the resulting skills-development dataset for remaining missing values after the sequential filters. The workflow does not implement a general duplicate-record check, date parsing, telephone-number validation, or an independent reconciliation of every monetary field with the source PDF.

The source PDF itself contains extraction challenges. Some tables continue across pages, some source values are split across lines, and some rows have missing fields. The skills-development section provides examples in which a value such as a term amount is split between lines in the PDF. The school-bursary section also contains a source record with a missing sex value, which the notebook manually assigns as M before applying its missing-value filter.

The final datasets therefore represent the cleaned output of the notebook rather than a direct export from the council's administrative database. Records removed by missing-value filters are not recoverable from the cleaned CSVs alone. Users who require a complete administrative record should retain the original PDF alongside the cleaned data and compare important records against the source.

5. Data Export and Reproducibility

The notebook uses pandas `to_csv` with `sep='|'` and `index=False` for the five cleaned DataFrames. The intended output names are `cleaned_community_projects.csv`, `cleand_loans_applicants.csv`, `cleaned_empowerment_grants.csv`, `cleaned_school_bursaries.csv` and `cleaned_skills_development_bursaries.csv`.

There are two output-code inconsistencies in the final notebook that should be corrected before treating the generated files as reproducible final deliverables. A later community-project save cell writes the `loans_applicants` DataFrame to `cleaned_community_projects.csv`, and the school-bursary save cell also writes `loans_applicants` to `cleaned_school_bursaries.csv`. The loan output filename is additionally spelled '`cleand_loans_applicants.csv`'. These statements do not change the cleaning operations performed on the in-memory DataFrames, but they can cause the files saved to disk to differ from the dataset names stated in the notebook. The paper therefore reports the final cleaned DataFrames produced by the cleaning logic and identifies the export issue explicitly.

6. Data Structure

The five datasets do not share a single record identifier. Ward is the most consistent geographical field and provides a basis for cross-dataset comparisons. The datasets should normally be analysed separately and linked only where the analytical question supports a common geographical or administrative comparison.

Dataset	Final records	Main subject	Key geographical fields
Community projects	7	Development projects	Ward, Zone, Location
Loan applicants	17	Loan applications and businesses	Ward, Zone
Empowerment grants	90	Income-generating projects	Ward
School bursaries	44	Secondary boarding education	Ward, School Location
Skills development bursaries	177	Skills and vocational training	Ward

7. Potential Reuse of the Data

The community-project data can be used to classify proposed projects by sector, project type, ward, zone and location. The loan data can support analysis of the businesses seeking CDF loans, their sectors and requested amounts. The empowerment data can be used to examine the distribution of approved activities and amounts across wards. The school-bursary data can support descriptive analysis of pupils by ward, sex, grade, school and fee requirements. The skills-development data can be used to examine programmes, training levels,

institutions, programme duration and tuition fields. Together, the datasets provide a structured basis for descriptive and exploratory analysis of different forms of recommended CDF support.

8. Data Accessibility

The source data are contained in the publicly accessible 2025 Gwembe Town Council PDF used for extraction. The processed datasets are intended to be distributed as pipe-delimited CSV files together with the Jupyter notebook used for extraction and cleaning. A persistent repository link and DOI should be added to the final manuscript once the cleaned datasets and notebook have been formally deposited.

9. Ethical and Data Handling Considerations

The source publication contains personal information, including applicant names and contact telephone numbers in the loan-application section. The processed data preserve information present in the source document. Users should avoid unnecessary redistribution or secondary use of personal contact information and should apply appropriate data-protection and repository requirements when sharing or analysing the records.

References

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